

Generation Interconnection System Impact Study Report

For Queue Project Q17-0784

Surrowiec 500 kV
December 02, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0784. The Interconnection Customer proposes a 488.7 MW solar photovoltaic facility interconnecting at the Surrowiec 500 kV substation in the Ohio area. The request was received on April 26, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 290 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,390,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: RAVENNA / OLD FORGE 230 kV line
Contingency: HYLEBOS / RAVENNA 500 kV Line
Loading: 108.4% (Rating: 864 MVA)
DFAX: 10.5%
Pre-project loading: 102.5%
2. Facility: IMPERIAL SUBSTATION / CHAMBERLIN 230 kV
Contingency: RAVENNA / HYLEBOS 500 kV
Loading: 108.4% (Rating: 857 MVA)
DFAX: 13.7%
Pre-project loading: 100.6%
3. Facility: MAPLE ST / ATASCOCITA 230 kV
Contingency: EL PASO / RAVENNA 500 kV
Loading: 101.8% (Rating: 563 MVA)
DFAX: 6.2%
Pre-project loading: 96.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
RAVENNA / OLD FORGE 230 kV line	Reconductor line	\$7,410,000	15% (\$1,092,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
IMPERIAL SUBSTATION / CHAMBERLIN 230 kV	Reconductor line	\$7,387,000	23% (\$1,684,000)
ATASCOCITA / MAPLE ST 230 kV Line	Reconductor line	\$16,105,000	16% (\$2,614,000)
	Total allocated to project		\$5,390,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.